In [ ]:

Data Loading and Exploration

I begin the project by importing the necessary

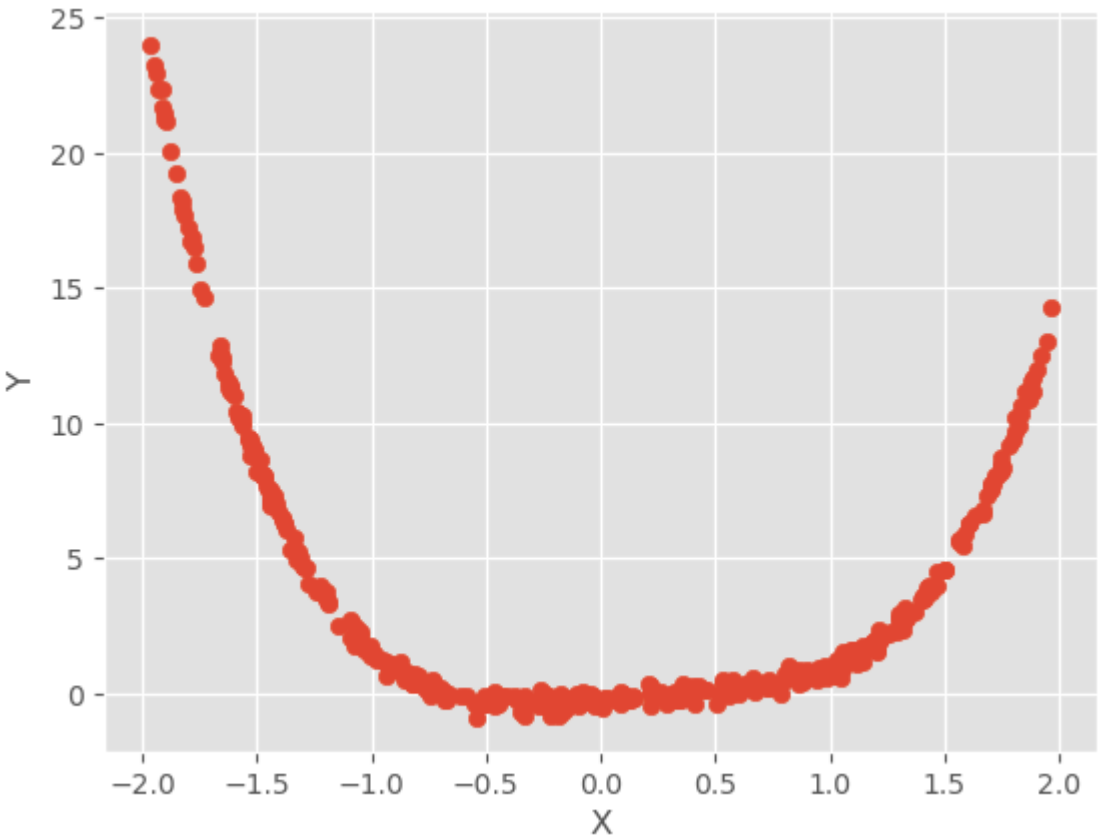
In [40]:
```
import matplotlib.pyplot as plt
import pandas as pd
import sklearn
from sklearn.preprocessing import PolynomialFeatures
import numpy as np
```

In [41]:
```
plt.style.use('ggplot')
```

In [42]:
```
train = pd.read_csv('train.csv')
train.head()
train.sort_values('x', inplace=True)

x = train.x.to_numpy()
y = train.y.to_numpy()
```

In [ ]:

In [43]:
```
plot1 = plt.scatter(x, y)
plt.xlabel('X')
plt.ylabel('Y')
plt.show()
```



In [44]:
```
poly = PolynomialFeatures(degree=4, include_bias=False)
poly_features = poly.fit_transform(x.reshape(-1, 1))
```

In [45]:
```
# poly.transform(x.reshape(-1, 1))
```

In [46]:
```
from sklearn.linear_model import LinearRegression

poly_reg_model = LinearRegression()
poly_reg_model
```

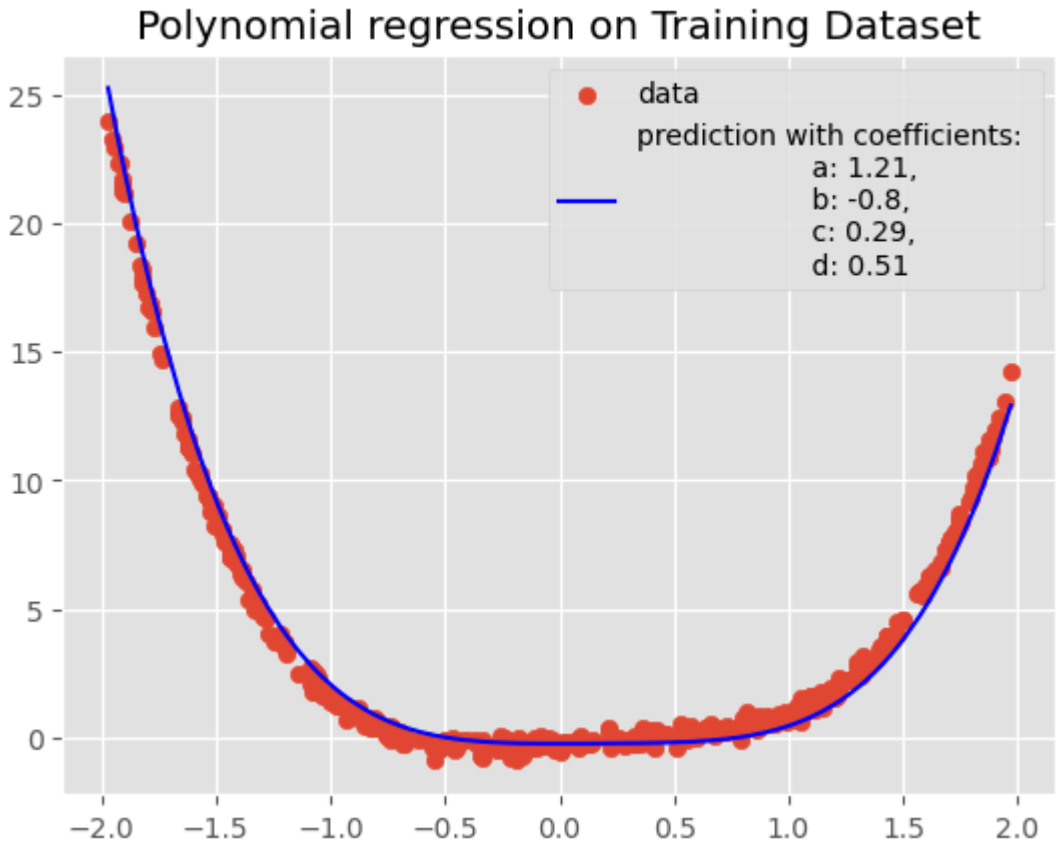
Out[46]:
```
▾ LinearRegression
LinearRegression()
```

In [47]:
```
params = poly_reg_model.fit(poly_features, y)

y_predicted = poly_reg_model.predict(poly_features)
# print(y_predicted[:10])
# print(y[:10])
```

In [48]:
```
d, c, b, a = params.coef_
print(a, b, c, d)
intercept = params.intercept_
print(params.intercept_)
```

```
1.2053934318561321 -0.8036620332156272 0.28989803950813753 0.5075934645790964
-0.21463817306568922
```

In [49]:
```
x_range = np.linspace(x.min(), x.max(), len(y_predicted))

x2 = np.sort(x)
y2 = np.sort(y)
pred = [a*x_i**4 + b*x_i**3 + c*x_i**2 + intercept for x_i in x2]

# plt.scatter(x, y)
a2, b2, c2, d2 = [round(i, 2) for i in [a, b, c, d]]
data = plt.scatter(x, y, label='data')
prediction = plt.plot(x2, pred, c='blue', label=f'''prediction with coefficients
                    a: {a2},
                    b: {b2},
                    c: {c2},
                    d: {d2}''')
plt.title('Polynomial regression on Training Dataset')
plt.legend()
plt.show()
# x_range
from sklearn.metrics import mean_squared_error

print(mean_squared_error(pred, y))

print(pred[0], y[0])
```

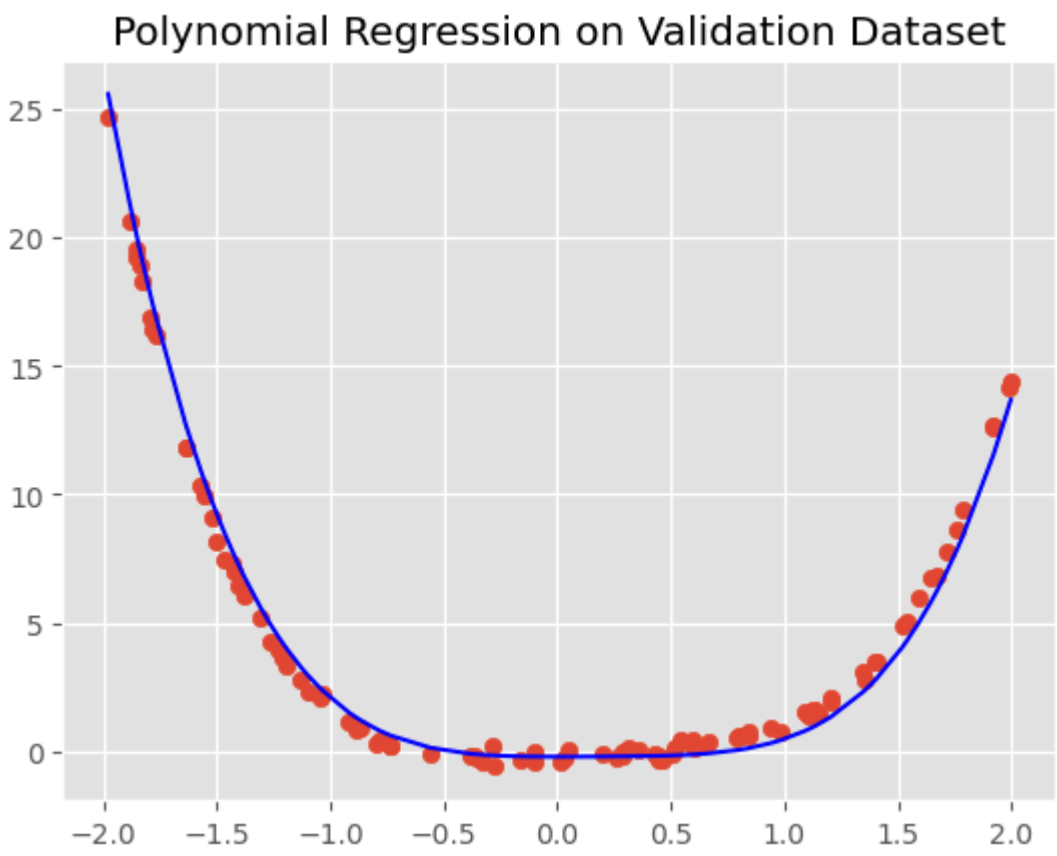



```
0.37636243729847907
25.235720081368484 23.98684764787348
```

In [50]:
```
validation_dataset = pd.read_csv('val.csv')

x_val = validation_dataset.x.to_numpy()
y_val = validation_dataset.y.to_numpy()
```

In [51]:
```
plt.scatter(x_val, y_val)
plt.plot(np.sort(x_val),
    [a*x_i**4 + b*x_i**3 + c*x_i**2 + intercept for x_i in np.sort(x_val)],
    c='blue')

plt.title('Polynomial Regression on Validation Dataset')
```

Out[51]:
```
Text(0.5, 1.0, 'Polynomial Regression on Validation Dataset')
```



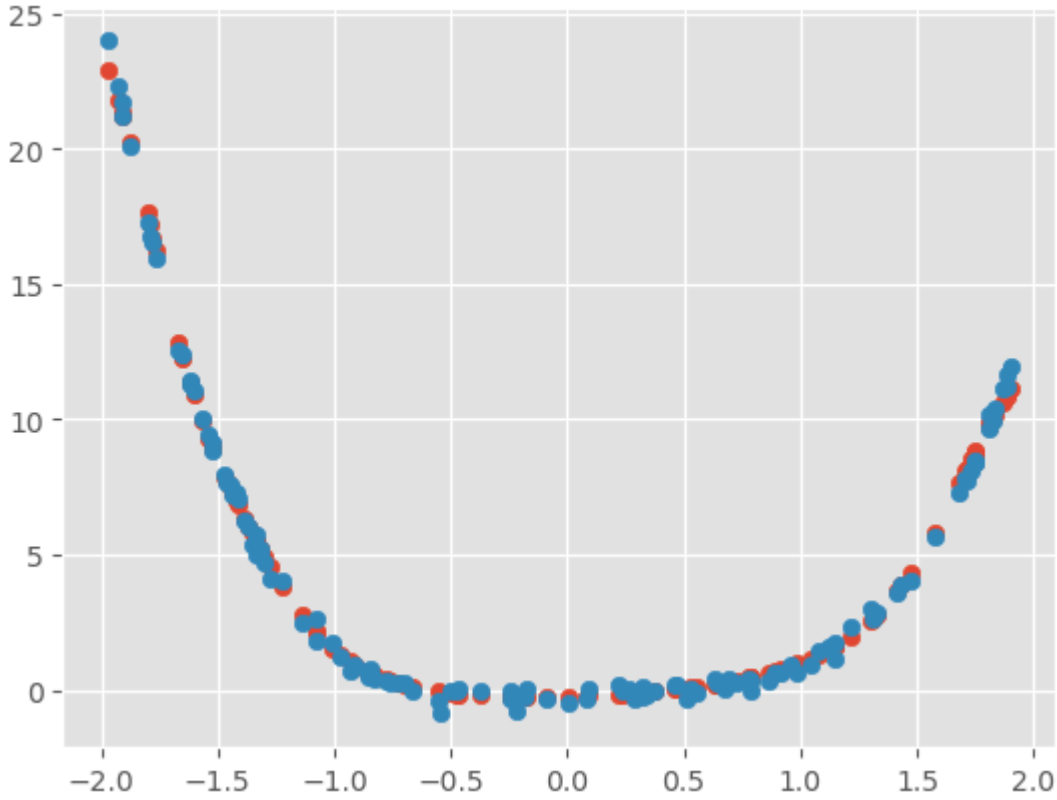
In [52]:
```
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPRegressor
from sklearn.metrics import mean_squared_error


x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.3, random_state=42
)

clf = MLPRegressor(activation='logistic', solver='adam', alpha=1e-5, learning_ra
                   learning_rate_init=0.001,
                   hidden_layer_sizes=(64, 64), random_state=1, max_iter=10_000)

clf.fit(x.reshape(-1, 1), y)
```

Out[52]:
```
MLPRegressor
MLPRegressor(activation='logistic', alpha=1e-05, hidden_layer_sizes=(6
4, 64),
             max_iter=10000, random_state=1)
```

In [53]:
```
y_true = y_test
y_pred  = clf.predict(x_test.reshape(-1, 1))
print(mean_squared_error(y_true, y_pred))
# print(y_test)
```

```
0.08541064028526574
```

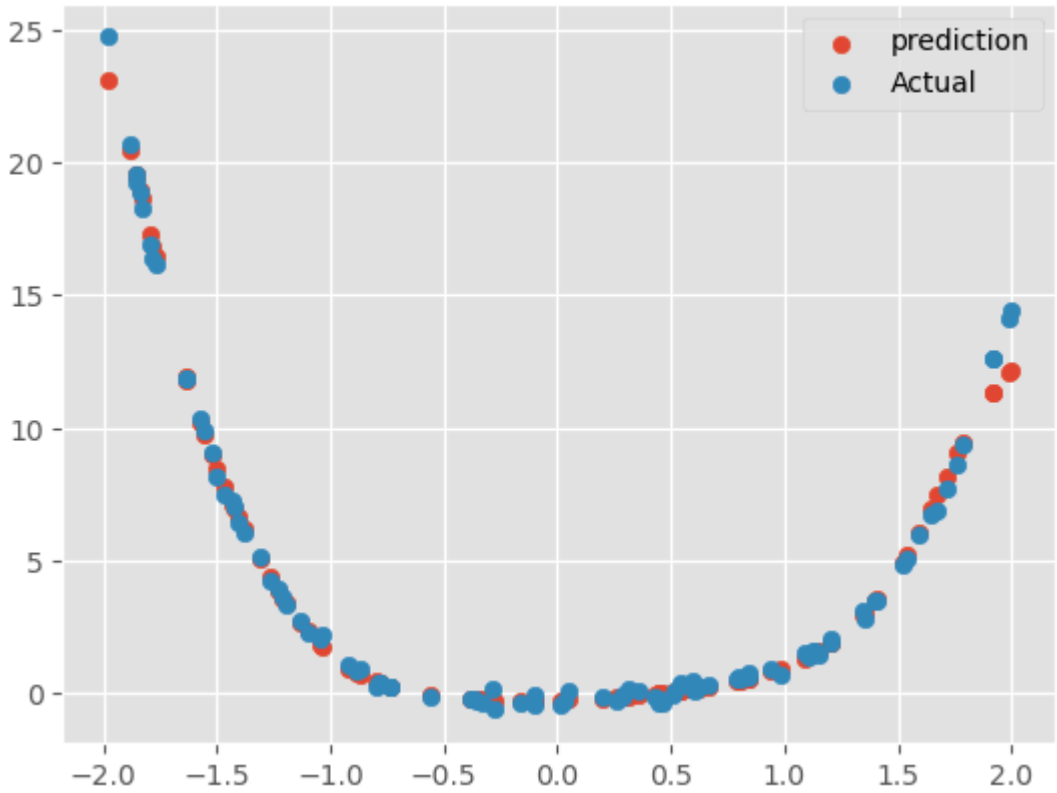
In [54]:
```
plot_3_1 = plt.scatter(x_test, y_pred)
plot_3_2 = plt.scatter(x_test, y_true)
```



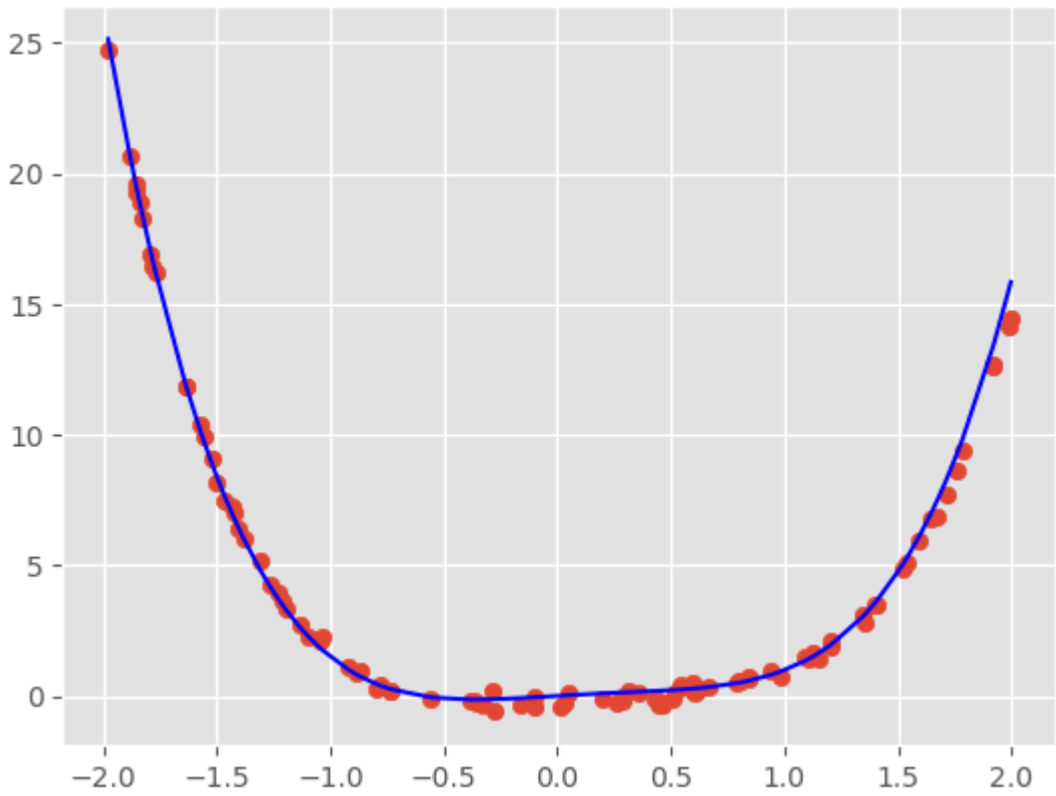
In [55]:
```
validation_dataset = pd.read_csv('val.csv')

x_val = validation_dataset.x.to_numpy()
y_val = validation_dataset.y.to_numpy()
```

In [56]:
```
y_val_pred = clf.predict(x_val.reshape(-1, 1))
```

In [57]:
```
plot_4_1 = plt.scatter(x_val, y_val_pred, label='prediction')
plot_4_2 = plt.scatter(x_val, y_val, label='Actual')
plt.legend()
plt.show()
```



In [58]:
```
plt.show()
```

In [59]:
```
A = [[i**4, i**3, i**2, i] for i in x_val]
Y = [i for i in y_val_pred]

res = np.linalg.inv(A[:4]).dot(Y[:4])
print(res)
x_val_2 = np.sort(x_val)
new_y_val_pred = [res[0]*i**4 + res[1]*i**3 + res[2]*i**2 + res[3]*i for i in x_
plt.plot(x_val_2, new_y_val_pred, c='blue')
plt.scatter(x_val, y_val, label='Actual')

print(y_val)
```

```
[ 1.34407042 -0.77210984 -0.10278485  0.51830356]
[ 3.80421612e-01  6.86078834e+00  1.61493147e+00  9.92923392e+00
  1.41354894e+01  4.85452891e+00  9.25548790e-01  3.09308445e+00
  1.03758256e+01  1.44102345e+01  3.27972390e-01  1.30712876e-01
  1.18503301e+01  5.95399585e+00  2.06868792e+01  2.24029875e+00
  7.27852390e+00  1.41634034e+00  3.65994997e+00  6.77369072e+00
  2.41800555e-01  1.92600241e+01  2.47229043e+01  1.68830314e+01
 -1.02327418e-01  1.89120964e+00  2.08233900e+00  3.48018999e+00
  1.61744091e+01  2.09540954e+00  7.73942265e+00  1.50349155e+00
  5.35561996e-01  2.80913208e+00  1.88991202e+01  3.31792680e+00
  8.17569903e+00 -3.99852851e-01  7.30285782e-01  1.26847359e-01
  3.49254837e+00  6.05359020e+00  9.27021003e-01  3.91451789e+00
  1.12216366e+00  6.36106897e-01  1.26562407e+01 -3.62283092e-01
  1.64164952e+01 -3.62831089e-01  1.82805866e+01  5.06296374e+00
  7.60442353e-01  5.19176657e+00  1.18445229e+01  4.25177795e+00
  1.26110073e+01 -3.75670281e-01  1.58218236e+00  3.35765369e-01
 -1.52552770e-01  7.01586826e+00  9.39722194e+00  2.81017518e-01
  1.67097263e-01  6.11931412e-01  2.07950790e-01  8.65789239e+00
  6.43570997e+00 -2.90520493e-01 -1.67902839e-01 -4.38565494e-01
  9.10721282e+00  7.47754034e+00  8.56496389e-01  4.42659944e-01
  1.97270198e-01 -2.17745450e-01  4.26645320e-01 -2.55380307e-01
 -2.00133741e-01  2.74740106e+00  8.64117513e-02  2.21809118e-01
  1.95409534e+01 -3.81094082e-01 -2.10522419e-02  2.28349497e+00
 -5.29433391e-02 -2.70903015e-01  5.80347350e-01  4.71366594e-01
  1.85439663e-01  2.23505192e-01  1.42949334e+00 -1.40262067e-01
 -6.09834091e-01 -7.93207900e-02 -8.57932781e-02  8.99623513e-02]
```